LLY Validation and AI Use Record

Validation and AI-use record

Validation results produced by the model

Run python lly_model.py; the generated output records these executable checks:

Requirement | Test | Expected result | Evidence

Synthetic / known-answer reconciliation | FY2025 balance sheet: $112,476m assets – $85,941m liabilities – $26,535m equity | Zero difference | `output/model-output.md`

Independent definition check | Debt/cash/share basis compared with the 2025 10-K balance sheet, Borrowings note, and EPS statement | Definitions and units disclosed; weighted-average-share limitation retained | `SOURCES.md`

Directional test written before execution | Higher WACC lowers per-share value; higher terminal growth raises it | Both pass | Model validation section

Three-statement articulation | Every forecast year: assets = liabilities + equity; cash from cash-flow statement = balance-sheet cash | Every test passes | Model validation section

Enterprise/equity bridge | (EV + cash – debt) / shares = reported per-share value | Pass | Model validation section

Peer policy / definition consistency | AMGN/GILD only; annual FY25 GAAP diluted EPS and date-matched prices; no debt/cash bridge to P/E | Qualified pass | `prior_work/Lab_08_comps.md`

Highest-risk failure test | Low case: slower volume, more price pressure, lower margins, higher capex/working capital, higher WACC | Low value is materially below base; action remains defer | `output/model-output.md`, `MEMO.md`

Assumption-challenge record

Load-bearing driver | Base value | Basis | Challenge / challenger | Evidence that would change it

Mounjaro/Zepbound volume | 32%, 28%, 22%, 17%, 12% | Decelerating path from FY25 franchise scale, not a straight-line extrapolation | Product concentration makes the terminal FCFF highly sensitive | Quarterly product volume / prescription and supply evidence materially above or below path

Realized price | -3%, -3%, -2%, -2%, -2% | Explicit pricing/rebate pressure, rather than assuming volume equals sales growth | Reimbursement/rebate compression can exceed low case | Quarterly net-price commentary, payer decisions, gross-to-net evidence

Gross margin | 83.0% to 81.0% | FY25 reported 83.0% then conservative normalization | Mix/manufacturing could help or pricing and capacity costs could hurt | Reported gross margin and manufacturing-cost trends

Capex | 13% to 9% of revenue | High early manufacturing investment, then normalization | Capacity expansion may not scale down on schedule | Reported PP&E purchases and capacity announcements

WACC / terminal growth | 8.5% / 3.0% | Analyst judgment and sensitivity bounds | Equity risk, capital structure, and terminal durability could invalidate it | Current market inputs and a documented cost-of-capital rebuild

Capital-allocation convention | Debt held at FY2025 level; distributions at 25% of NI | Transparent convention needed to articulate statements | Not a forecast of management financing | Updated cash, debt, dividend, buyback, and financing disclosures

Locked Changed-Input Record

The prediction below was saved before the documented rerun. No Git client is installed in this workspace, so the timestamp and this frozen written record serve as the pre-run evidence.

Precommit timestamp | Commit / frozen copy | Input and units | Old → new | Predicted direction | Predicted decision effect

2026-10-19 13:55:22 EDT | `VALIDATION_AND_AI_USE.md` saved before rerun; no Git client available | Base WACC; percent | 8.5% → 9.5% | Base per-share value will fall materially because both explicit FCFF and terminal value are discounted more heavily. | No change: Watch/Defer should remain because even the original base value is below the preserved market reference.

Actual before/after | Actual decision effect | Prediction reconciliation | Failure diagnosis or no-change reason | Evidence path

Base DCF per-share value: $637.42 at 8.5%; $529.24 at 9.5%; change: -$108.18 (-17.0%). Terminal-value share fell from 80.5% to 77.4%. | No change: Watch/Defer. $529.24 remains below the preserved $1,124.21 market reference. | Correct. Value fell materially and the decision did not change. | No failure: the outcome follows the pre-stated DCF discount-rate direction. The case still requires monitoring because the market-price comparison uses later market data than FY2025 bridge inputs. | `output/changed-input-wacc-9_5.md`; restored base output in `output/model-output.md`; original 8.5% input restored in `lly_model.py`.

AI output check

AI output considered | Independent check | Disposition

Proposed integrated-model architecture and constant-debt equation | Algebraic cash-flow-to-cash and balance-sheet checks in `lly_model.py`; 10-K source map checked against filing locations | Accepted with qualification: constant debt is disclosed as a model convention, not a company forecast.

Earlier simple growth-only DCF | Replaced by product-driver forecast and articulated statements | Rejected as final architecture; preserved in `prior_work/` for evolution evidence.

Partner cold run

Operator: Independent partner, not the project author. AI used during the run was the partner's, not the project author's. This first attempt exposed an environment/package-access failure before the model could execute.

Run date/time | Operator / AI use | OS | Python version | Elapsed time to visible output | First error or `none` | Outcome

Date not supplied by partner | Independent partner; Codex used only to conduct/document attempt, with no model modification | Windows | `python --version` failed: Python not recognized/on PATH | ~8 seconds to first environment error | `python : The term 'python' is not recognized as the name of a cmdlet, function, script file, or operable program.` Prior file search also found no `lly_model.py` or `README.md` in the active folder. | Failed before model execution. `output/model-output.md` was not created.

Repair / clean rerun required: Put README.md and lly_model.py together in one new folder on the partner's computer, install Python 3 and select Add Python to PATH, then have the partner run python --version and python lly_model.py without coaching. Record that second, clean result separately; it is required before claiming a successful cold run.

Clean-rerun date/time | Operator / AI use | OS | Python version | Elapsed time to visible output | First error or `none` | Outcome

Date/time not supplied by partner | Independent partner; Codex used by the partner during the rerun | Windows (reported in first attempt) | Python 3 (major version reported) | 2 minutes 34 seconds | none | Passed. `output/model-output.md` was created.

The three recordings, corrected transcripts, repository access check, and final receipt still require student action.